GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Pig Trap Calculation Note**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-006
Document Revision	R01
Document Status	Issued for Review
Originator	Joseph Obinna
Security Classification	Restricted
Issue Date	27-Jul-22

Pig Trap Calculation Note

R01	27.07.22	Issued for Review	J. Obinna	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	27.07.22	Issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.

b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)

for the subsequent project phases, post FEED, it is expected that:

c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.

d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)

e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-006

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001	Specification for Line pipe
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-011	Specification for Pig Traps
	FISAS Draft ESIA Report (November 2020)

Table of Contents

Abbreviations

Abbreviation	Meaning
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
AWS	American Welding Society
CS	Carbon Steel
DP	Design Pressure
ISO	International Organisation for Standardization
MMscf/d	Million standard cubic feet per day
OD	Outer Diameter
PE	Polyethylene
PSL	Product Specification Level
TD	Design Temperature

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

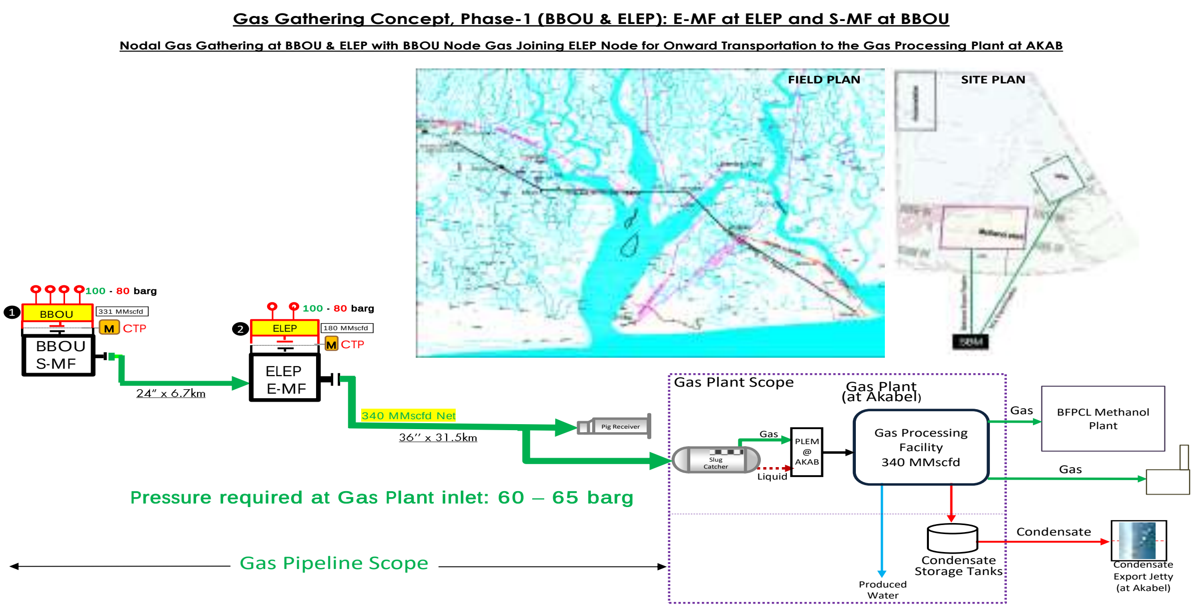

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of the document is to determine the wall thicknesses of the Major and Minor barrels to be used in the design of the Pig Traps of the 24″ BBOU to ELEPA pipeline system and the 36″ ELEPA to AKABEL Pipeline system. The design is based on material API 5L grade X70 and in accordance with industry standard codes ASME B31.8.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
ASME B31.8	Gas Transmission and Distribution Piping Systems
API 5L	Specification for Line Pipe
ASME B36.10M	Welded and Seamless Wrought Steel Pipe
ISO 3183: 2007	Petroleum and Natural Gas Industries Steel Pipe for pipeline transportation systems

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Pipeline Calculation Basis

3.1. Location Class/Design Factor

The entire pipeline systems including the Pig traps are designated for location class 1. However, the wall thickness of the pipeline system was calculated with a design factor of 0.6 (Used for Location class 2) for the entire stretch of the Pipeline land and swamp Route. This is to mitigate against the usual occurrence in Nigeria where overtimes, new habitats spring up along pipeline ROW.

The calculation of the pipe wall thicknesses to be used for the major and minor barrels of the pig traps shall be carried out using the same principle for the pipeline wall thickness calculation as per ASME B31.8.

Where a nonstandard wall thickness is calculated after adding the corrosion allowance, The next standard wall thickness as per ASME B36.10M shall be recommended.

The selected pipe shall have a diameter to wall thickness (D/t) ratio that is not greater than 96

3.2. Hoop Stress Design Factors

The wall thickness shall be capable of withstanding the load exerted on the wall of the barrels pipeline by the fluid internal pressure. The resulting circumferential (hoop) stress shall not exceed the specified minimum yield stress of the pipe material, multiplied by a factor applicable to the installation location. The corrosion allowance was added to the calculated wall thickness to determine the nominal wall thickness.

3.3. Hydrotest Pressure.

The required wall thickness for hydrotest shall be 1.25 x DP Per Table 841.3.2-1 of ASME B31.8

3.4. Corrosion Allowance.

The corrosion allowance for the 24" Pipeline system pig traps located at BBOU and ELEPA Manifold Platforms respectively is 3mm while the corrosion allowance of the 36" Pipeline system located at ELEPA and AKABEL is 6mm

3.5. 24" Multiphase Pipeline Data

The pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L – X70(PLS2)
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	24"
Density	7,850 kg/m3
Modulus of elasticity	2.07 x 10^5 MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10-6 m/ m/ ºC
Ultimate Tensile Strength	570 MPa
Steel Specific Resistivity	0.018 Ω m/mm2
OD / Wall Thickness	15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	60mm
Installation Condition	Trenching
Cathodic Protection	Impressed current
Design Life	30 years

3.6. 36" Multiphase Pipeline Design Data

The pipeline design data for the Multiphase Pipeline Project are as shown below

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L – X70 (PLS2)
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	36"
Density	7,850 kg/m3
Modulus of elasticity	2.07 x 10^5 MPa
Poisson's ratio	0.3
Coefficient of linear thermal expan-	11.7 x 10-6 m/ m/ ºC
Ultimate Tensile Strength	570MPa
Steel Specific Resistivity	0.018 Ω m/mm2
OD / Wall Thickness	23.83mm
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	85mm
Installation Condition	Trenching
Cathodic Protection	Impressed current
Design Life	30 years

3.7. Pipeline Material

The material of line pipe to be used for the pig traps major and minor barrels is the same material used for the mainline pipes; carbon steel material grade of API-5L GR X70 with Product Specification Level - PSL 2.

4. Major and Minor Barrel Wall Thickness Calculation

4.1. Calculation Procedure

For determination of the required wall thickness on the barrels at the design pressure, the following equations were used in accordance with ASME B31.8.

to = P * D / 2 * F * S * E * TEq. 1

The nominal thickness of straight section of a steel pipe is given by equation 2:

tn = to + CA Eq. 2

Where:

P =Internal design pressure (barg)

tn =Nominal wall thickness (mm)

to =Pressure Containment Wall Thickness (mm)

CA =Corrosion Allowance (mm)

D =Outside diameter of pipe (mm)

S =Specified minimum yield strength of the pipe (mm HG)

E =Longitudinal weld joint factor

F =Design factor

T =Temperature derating factor

4.2. Summary and Recommendation

The wall thickness was determined to satisfy the requirements of ASME B 31.8a Section 841.11. Based on the calculated minimum wall thickness, the recommended standard wall thickness was selected from ASME B 36.10 and are shown in the table below. The standard size selected listed below includes the corrosion allowances.

Appendix 1 to 4 are the spread sheet calculation used for the wall thickness calculations.

WALL THICKNESS CALCULATION FOR MINOR AND MAJOR PIG TRAPS									
Equip-ment / Tag.	Location	Barrel Type	Material	Nominal Pipe Size	OD (mm)	Design Factor	Calcu-lated WT (mm)	Selected WT (mm)	Location Class
Pig Launcher /	BBOU MS	Minor	API-5L GR X70	24″	610	0.6	14.58	15.88	1

BBOU-A-200		Major	API-5L GR X70	28″	711	0.6	16.49	17.48	1
Pig Receiver / ELEP-A-200	ELEPA MS	Minor	API-5L GR X70	24″	610	0.6	14.58	15.88	1
		Major	API-5L GR X70	28″	711	0.6	16.49	17.48	1
Pig Launcher / ELEP- A-300	ELEPA MS	Minor	API-5L GR X70	36″	914	0.6	23.35	23.83	1
		Major	API-5L GR X70	40″	1016	0.6	25.28	25.48	1
Pig Receiver / AKAB-A-100	AKABEL PS	Minor	API-5L GR X70	36″	914	0.6	23.35	23.83	1
		Major	API-5L GR X70	40″	1016	0.6	25.28	25.48	1

Legend:

OD – Outside Diameter

WT – Wall Thickness

MS – Metering Station

PS – Pigging Station

Appendix 1: 24" Minor Barrel Wt Calculation (According to ASME B 31.8)

Appendix 2: 28" Major Barrel Wt Calculation (According to ASME B 31.8)

Appendix 3: 36" Minor Barrel Wt Calculation (According to ASME B 31.8)

Appendix 4: 40" Major Barrel Wt Calculation (According to ASME B 31.8)

APPENDIX 1: 24" MINOR BARREL WT CALCULATION (ACCORDING TO ASME B 31.8)

Location Class 1(Designated), Design Factor 0.6:

Nominal wall thickness		t_m = PD/(2S FET)		(As Per 841.1.1 of ASME B31.8)		
DP -	Design pressure (110 barg)		=	11	MPa
DT -	Design Temperature		=	60	^{o}C
D -	Outside diameter	(24")	=	610	mm
S	SMYS of pipe material- API 5L GR X70 (See Table D-1)		=	483	MPa
F -	Design factor as per table 841.1.6 (For class 1- Designated)		=	0.6	
E -	Longitudinal joint factor as per table 841.17-1 (For SAW Pipe)		=	1	
T -	Temperature derating factor as per table 841.1.8-1 (For Design temp = 60 ^{o}C)		=	1	
t_m -	Nominal wall thickness		=	11.58	mm
CA -	Corrosion allowance		=	3	mm
t_C -	Calculated wall thickness	t_m + CA	=	14.58	mm
Selected Regualar standard wall thickness as per API 5L (Table 9) and AMSE B36.10M		t	=	15.88	mm
Internal diameter		D-2t	=	578.24	mm

Wall Thickness check for Hydrotest

$(P_H D/2t_{C)} + P_h < 90\%$ SMYS — 264.14 = 54.69 %SMY! OK

P_H	-	Test Pressure, MPa (1.25 x DP) Per Table 841.3.2-1	13.75	Mpa
		$(P_H D/2t)$	264.09	MPa
P_h	-	Hydrostatic Pressure. = ρgH	0.05	MPa
D	-	Nominal Diameter, mm		
t	-	Wall Thickness, mm		
ρ	–	Density of water	1000	kg/m^3
g	–	Gravity	9.8	m/S^2
H	–	Height (maximum difference of elevation along the ROW) (m)	5	m

APPENDIX 2: 28" MAJOR BARREL WT CALCULATION (ACCORDING TO ASME B 31.8)

Location Class 1(Designated), Design Factor 0.6:

Nominal wall thickness			t_m = PD/(2S FET)		(As Per 841.1.1 of ASME B31.8)	
DP	-	Design pressure (110 barg)		=	11	MPa
DT	-	Design Temperature		=	60	^{o}C
D	-	Outside diameter	(28")	=	711	mm
S		SMYS of pipe material- API 5L GR X70 (See Table D-1)		=	483	MPa
F	-	Design factor as per table 841.1.6 (For class 1- Designated)		=	0.6	
E	-	Longitudinal joint factor as per table 841.17-1 (For SAW Pipe)		=	1	
T	-	Temperature derating factor as per table 841.1.8-1 (For Design temp = 60 ^{o}C)		=	1	
t_m	-	Nominal wall thickness		=	13.49	mm
CA	-	Corrosion allowance		=	3	mm
t_C	-	Calculated wall thickness	t_m + CA	=	16.49	mm
Selected Regualar standard wall thickness as per API 5L (Table 9) and AMSE B36.10M			t	=	17.48	mm
Internal diameter			D-2t	=	676.04	mm

Wall Thickness check for Hydrotest

$(P_H D/2t_{C)} + P_h < 90\%$ SMYS — 279.69 = 57.91 %SMY! OK

P_H	-	Test Pressure, MPa (1.25 x DP) Per Table 841.3.2-1		13.75	Mpa
		$(P_H D/2t)$	=	279.641	MPa
P_h	-	Hydrostatic Pressure. = ρgH		0.05	MPa
D	-	Nominal Diameter, mm			
t	-	Wall Thickness, mm			
ρ	–	Density of water	=	1000	kg/m^3
g	–	Gravity	=	9.8	m/S^2
H	–	Height (maximum difference of elevation along the ROW) (m)	=	5	m

APPENDIX 3: 36" MINOR BARREL WT CALCULATION (ACCORDING TO ASME B 31.8)

Location Class 1(Designated), Design Factor 0.6:

Nominal wall thickness		t_m = PD/(2S FET)		(As Per 841.1.1 of ASME B31.8)	
DP -	Design pressure (110 barg)		=	11	MPa
DT -	Design Temperature		=	60	^oC
D -	Outside diameter	(36")	=	914	mm
S	SMYS of pipe material- API 5L GR X70 (See Table D-1)		=	483	MPa
F -	Design factor as per table 841.1.6 (For class 1-Designated)		=	0.6	
E -	Longitudinal joint factor as per table 841.17-1 (For SAW Pipe)		=	1	
T -	Temperature derating factor as per table 841.1.8-1 (For Design temp = 60 ^oC)		=	1	
t_m -	Nominal wall thickness		=	17.35	mm
CA -	Corrosion allowance		=	6	mm
t_C -	Calculated wall thickness	t_m + CA	=	23.35	mm
Selected Regualar standard wall thickness as per API 5L (Table 9) and AMSE B36.10M		t	=	23.83	mm
Internal diameter		D-2t	=	866.34	mm

Wall Thickness check for Hydrotest

$(P_H D/2t_{C)} + P_h < 90\%$ SMYS — 263.74 = 54.60 %SMY: OK

P_H	-	Test Pressure, MPa (1.25 x DP) Per Table 841.3.2-1	13.75	Mpa
		$(P_H D/2t)$	263.691	MPa
P_h	-	Hydrostatic Pressure. = ρgH	0.05	MPa
D	-	Nominal Diameter, mm		
t	-	Wall Thickness, mm		
ρ	–	Density of water	1000	kg/m^3
g	–	Gravity	9.8	m/S^2
H	–	Height (maximum difference of elevation along the ROW) (m)	5	m

APPENDIX 4: 40" MAJOR BARREL WT CALCULATION (ACCORDING TO ASME B 31.8)

Location Class 1(Designated), Design Factor 0.6:

Nominal wall thickness — t_m = PD/(2S FET) — (As Per 841.1.1 of ASME B31.8)

DP -	Design pressure (110 barg)		=	11	MPa
DT -	Design Temperature		=	60	^{o}C
D -	Outside diameter	(40")	=	1016	mm
S	SMYS of pipe material- API 5L GR X70 (See Table D-1)		=	483	MPa
F -	Design factor as per table 841.1.6 (For class 1-Designated)		=	0.6	
E -	Longitudinal joint factor as per table 841.17-1 (For SAW Pipe)		=	1	
T -	Temperature derating factor as per table 841.1.8-1 (For Design temp = 60 ^{O}C)		=	1	
t_m -	Nominal wall thickness		=	19.28	mm
CA -	Corrosion allowance		=	6	mm
t_C -	Calculated wall thickness	t_m + CA	=	25.28	mm
Selected Regualar standard wall thickness as per API 5L (Table 9) and AMSE B36.10M		t	=	25.40	mm
Internal diameter		D-2t	=	965.20	mm

Wall Thickness check for Hydrotest

$(P_H D/2t_{C)} + P_h < 90\%$ SMYS — 275.05 = 56.95 %SMY: OK

P_H	-	Test Pressure, MPa (1.25 x DP) Per Table 841.3.2-1		13.75	Mpa
		$(P_H D/2t)$	=	275	MPa
P_h	-	Hydrostatic Pressure. = ρgH		0.05	MPa
D	-	Nominal Diameter, mm			
t	-	Wall Thickness, mm			
ρ	–	Density of water	=	1000	kg/m^3
g	–	Gravity	=	9.8	m/S^2
H	–	Height (maximum difference of elevation along the ROW) (m)	=	5	m